

SRW3 Phase 1I-R2 — Gate-Verdict Provenance Binding Report

Branch: phase1i-r2-gate-provenance (additive tree phase1i-r2/ only)

Base: 0b667af79659b783f813a416aad84d641b168034 (= tag phase1i-r1-complete = branch phase1i-r1-formal-binding)

Backend evidence: K Haskell (proofs) + K LLVM with the pinned krypto shim (concrete keccak) + Python executable mirror (real keccak)

Companion documents: phase1i-r2/audit/GATE-PROVENANCE-AUDIT.md (defect record, isolation analysis, correspondence discipline)

1. Executive summary

Phase 1I-R1 closed the accept-side trust boundary: the accept command no longer takes a verdict or a configuration, and only machine-created receipts can authorize a commit. But R1's GATE command still had the shape `pGateEval(ProtoBlock, Bytes, Bytes, String)` — the caller supplied the verdict `STRING`, the mint stored it verbatim, and no rule of the R1 machine ever evaluated the frozen gate. A caller that typed `"valid-g"` obtained a receipt claiming the frozen gate accepted. This report closes that mint-side boundary.

The repair is an **isolated R2 machine** (`srw3proto-r2.k`) that imports **ONLY** the frozen Phase 1G gate module `SRW3AUTHZ` — not `SRW3PROTO`, not `SRW3PROTO-R1` — so the legacy commands and transitions are not merely avoided but **ABSENT** from the machine's command language and rule set. In it:

- the gate command `pGateEvalR2(B, FC, RG, CERT, ERC, ERP)` takes evidence objects and client-derived execution results — and ****no verdict argument of any sort****;
- the machine **constructs** the authority context from the pinned configuration (authorized `SecurityContext` + pinned proof-type/guest/config parameters), **supplies** the position (slot counter + lineage anchor), and **computes** the receipt verdict as `VerifyLineageG(RG, AC, T, HEAD)` — the frozen 21-conjunct gate is the only producer of `"valid-g"`;
- the named predicate `GateInputMatchesBlockR2` (plus `GateAuthorityMatchesCfgR2`, `GateCtxAuthorizedR2`) binds every presented evidence object to the EXACT candidate and the pinned configuration;
- the accept consumes the receipt once, requires the DERIVED decision of the COMPUTED verdict to be `pdecValidR2`, and re-checks the full commitment predicate against the pin.

Results. K proof ladder: **37 PROVED** (including the headline R2-ACCEPT-IMPLIES-COMMITP, the mint-shape pair, the full rejection family, the correspondence claims to the frozen 1I formulas, and the ported frozen 1I ladder) and **3 NOT MECHANIZED** (the two full-pipeline compositions and the universal closure — each with the exact stuck log retained, and each with its concrete substitute demonstrated). Concrete gate evaluation is demonstrated on the LLVM backend with real keccak via the pinned krypto shim: the honest mint computes `gr2Verdict: "valid-g"`, the honest two-block pipeline commits, and every attack computes a rejection or is ill-formed. The Python mirror (real keccak, the frozen 1G gate imported unchanged) passes **87/87**, including the byte-level cross-layer certificate that reproduces the K digests digit-for-digit and anchors to the FROZEN 1G transcript (`certid0 = 89ffea58...`). Zero regression: 11/11, 26/26, 30/30, 15/15, 17/17, 16/16, 48/48 — exact.

Scientific verdict: CLOSED for the in-model gate-verdict-provenance question, with the standing boundaries restated (section 8) — R2-ALL remains NOT MECHANIZED, the Level-III model boundary is unchanged, and no consensus-integration claim is made.

2. Starting point and ground rules

The repair starts from `0b667af` (tag `phase1i-r1-complete`) and works in an additive `phase1i-r2/ tree`. `phase1g`, `phase1h`, `phase1h-r1-fix`, `phase1i`, and `phase1i-r1-formal-binding` are preserved byte-for-byte (no amend, no rebase, no force-push; verified by the frozen-ref check in section 9 and the provenance record). The K toolchain was rebuilt from the pinned recipe (K v7.1.337, z3 4.13.3, the clang/LLVM-15 chain at 15.0.6-4+b1, the Phase-1D krypto shim at `k/phase1d/shim` with the multiblock fix, `blockchain-k-plugin` K sources at `207ae512`); the rebuild is scripted (`scripts/r2_rebuild_toolchain.sh`, `scripts/r2_build_llvm.sh`) and logged.

3. The audit (what was wrong, mechanically witnessed)

Finding **R2-F1**: the R1 mint stores the caller's verdict string verbatim; no R1 rule evaluates `VerifyLineageG`. Three independent witnesses are retained against the UNMODIFIED R1 module:

1. **K witness claims** (`old_r1_witness.k`, `kprove` over the R1 definition):
`OLD-R1-GATE-MINTS-CALLER-VERDICT`, `OLD-R1-ACCEPTS-CALLER-VERDICT`
 (the accept fires and the chain extends — the security equalities appear as hypotheses precisely because nothing in R1 could verify or refute

them), OLD-R1-PIPELINE-FORGED. All three WITNESSED.

2. **Executable witness** (Python, real keccak): the unmodified R1 logic accepts after `gate_eval(..., "valid-g")`; the receipt verdict is captured as the caller's text (INJ-R1-* cases, 87-case suite).
3. **K run transcripts** on the R1 definition showing the injected verdict inside the minted receipt (transcripts/legacy/).

The full defect record with line references is

`audit/GATE-PROVENANCE-AUDIT.md §A`.

4. The repair design

Isolated machine surface. SRW3PROTO-R2 imports SRW3AUTHZ only. The legacy commands have no production (kparse rejects every legacy name — transcripts T9/T10a-d) and the legacy transitions are not in the rule set (audit §B rule inventory). This is the handoff's preferred architecture in its strongest form: an adversary controlling the admitted R2 command stream cannot reach the old unsafe transition because it does not exist in this machine.

Computed verdict. `pGateEvalR2` constructs

``AC_m = azCtx(FC, ProtoAuthorizedSecCtxR2(C, pbParentRootOf(B)), CERT, pcAuthPTOf(C), pcAuthGuestDOf(C), pcAuthCfgDOf(C))`` — the caller cannot name the `SecCtx` or any authority parameter — and writes `gr2Verdict := VerifyLineageG(RG, AC_m, N, LH)` with `N` the machine slot counter and `LH` the machine lineage anchor. The mint is the only receipt producer (audit §C); no command carries a verdict, a receipt, or a replacement configuration.

Candidate/context binding. `GateInputMatchesBlockR2(FC, RG, CERT, B)` (payload digest via the certificate field + the frozen L28 hash closure; parent state root; post-state root; effect-trace digest; evidence identity via the record's certificate digest + the frozen L28 `AzCertId` recomputation) and `GateAuthorityMatchesCfgR2` (pinned schedule/spec) guard the mint; the accept re-checks them on the receipt-stored inputs. The policy-commitment and context-digest equalities are checked directly against the pin at accept. The handoff §5.2 checklist is discharged item-by-item in audit §F.

Pinned configuration. `ProtoCfgR2` extends the 1I config shape with the gate-authority fields 9..13 (sched/spec/authPT/authGuestD/authCfgD — the handoff §5.3 requirement). `pInitR2` pins once (reserved key `-1`), refuses re-initialization, and anchors the protocol head at the R2 protocol root and the lineage head at the 32-zero-byte lineage root. There is deliberately no

upgrade transition.

Two chain heads. The protocol chain (1I, head := 0x9E block commitment) and the evidence lineage chain (1D/1G, records anchor at the parent's F-child) are distinct frozen chains; R2 tracks both (<r2head>, <r2linhead>), the gate's HEAD is the lineage anchor, receipt freshness is enforced on both, and acceptance advances the lineage head to the accepted record's F-child (R2RecChildF).

Receipt. An R2GateReceipt binds the exact candidate, the client execution results, the exact gate inputs (FC/RG/CERT/AC), the COMPUTED verdict, head/slot/lineage-head at evaluation time, and the pinned config object. Missing, duplicate, stale, mismatched, or rejected receipts fail closed with state unchanged.

5. The K proof ladder (37 PROVED / 3 NOT MECHANIZED)

All on the Haskell backend over the R2 definition (run_r2_proofs.sh; logs in transcripts/k/). The headline:

Claim	Status	Content
R2-ACCEPT-IMPLIES-COMMITP	PROVED	the accept guard verbatim $\Rightarrow$ CommitPR2(B, ERC, ERP, C, pbParentRootOf(B)); Commit_P only in the destination; the decision conjunct is the DERIVED decision of the computed verdict
R2-VALID-ACCEPT	PROVED	non-vacuity: under the same guard the transition fires and extends the chain
R2-MINT-SHAPE-EXISTING-POOL / -EMPTY-POOL	PROVED	the mint writes the computed verdict + the bound inputs (the derivation chain of audit §C)
R2-CONFIG-FRAME-GATE / -ACCEPT	PROVED	the pinned config is never mutated by gate/accept
R2-INIT-PINS / R2-CONFIG-IMMUTABLE	PROVED	one-time pin; re-init refuses
R2-DECISION-BOUND	PROVED	the accepted decision field equals the gate-derived decision
R2-GATELINK-VALID / -REJECT1..3	PROVED	only "valid-g" maps to pdecValidR2; rejection verdicts map to pdecRejectR2 (fail-closed)
R2-FORGED-VERDICT-REJECT / -AUTHTYPE	PROVED	computed rejection verdicts cannot accept (state unchanged)
R2-REJECT-NO-POOL / -EMPTY-POOL / -WRONG-CANDIDATE / -STALE-HEAD / -STALE-LINHEAD / -WRONG-SLOT / -GATE-INPUT-MISMATCH / -CONFIG-SUB	PROVED	the fail-closed family (concrete countermodels)
R2-RECEIPT-SINGLE-USE	PROVED	one receipt, one commit; replay refused
PI1/PI2/PI5/PI7 + PI2-EXEC-INVALID (ported)	PROVED	the frozen 1I ladder over the R2 predicate forms
R2-CORRESPONDENCE-CONFIG-ROOT / -POLICY-COMMITMENT / -CTX-DIGEST / -BLOCK-COMMIT	PROVED	the R2 predicates equal the FROZEN 1I formulas (transcribed byte-for-byte; audit §D)
R2-NO-CALLER-VERDICT	PROVED (syntactic)	no accepted command has a verdict/decision argument — the sort audit + kparse rejection transcripts T9/T10a-d

Claim	Status	Content
R2 - GATE - RESULT - DERIVED	PROVED (composition)	mint-shape claims + producer exhaustiveness (audit §C)
R2 - GATE - INPUT - BINDING	PROVED (composition)	mint guard + stored-input re-checks
R2 - INVALID - CANDIDATE - REJECT	PROVED	R2 - FORGED - VERDICT - REJECT + the VERDICT family + LLVM/Python
R2 - FORGED - VERDICT - IMPOSSIBLE	PROVED (surface) + WITNESSED	the injection is ill-formed (T9) ; the R1-side injection formally witnessed (OLD-R1 - *)
R2 - LEGACY - SURFACE - EXCLUDED	PROVED (structural)	audit §B: no legacy productions, no legacy rules; parse transcripts
R2 - PIPELINE - HONEST	NOT MECHANIZED	hs stuck-term boundary: the accept guard must decide the derived decision over the (unevaluated) computed verdict; exact log retained
R2 - PIPELINE - FORGED	NOT MECHANIZED	same boundary
R2 - ALL	NOT MECHANIZED	the universal closure (circularity claim) — configuration cannot be rewritten further; log retained

Substitutes for the three NOT-MECHANIZED items (each disclosed, none silent): the full honest pipeline is CONCRETELY demonstrated on the LLVM backend (T2: computed verdict "valid-g" → commit; T3: two-block continuation; T6: the forged end-to-end — computed rejection → state unchanged) and in the Python mirror (HONEST/HONEST2 + INJ + VERDICT groups); the K-side composition is carried by the PROVED halves (mint-shape $\rightleftharpoons$ accept-side), which is the composition identity for the two steps; R2 - ALL remains open exactly as in R1/1I (the same class of kore blocker, log retained).

6. Concrete gate evaluation (LLVM + krypto shim) and the demo suite

The pinned shim (k/phase1d/shim, multiblock fix marker-checked) gives the LLVM backend real keccak-256 and secp256k1 — the same executable path as the frozen 1G demo suite. transcripts/demos/ (LLVM, run_demos.sh):

Demo	Result
T1 gate eval	receipt minted with gr2Verdict: "valid-g" COMPUTED over the bound evidence; no chain state moves
T2 honest accept	COMMITTS slot 0; head := block commitment; lineage head := record F-child
T3 two-block continuation	slots 0 and 1 both commit; the lineage head threads the F-child into the slot-1 gate position
T4 accept, no receipt	clean reject, state unchanged
T5 re-initialization	refused, state unchanged
T6 config-version substitution	reject, state unchanged (CV=2 vs pinned 1)
T7 position substitution	mint refuses (slot binding); accept finds no receipt

Demo	Result
T8 evidence/candidate mismatch	mint refuses (GateInputMatchesBlockR2)
T9 the R1 injection pGateEvalR2(. . . , "valid-g")	ILL-FORMED (parse error; no such production)
T10a-d legacy pAccept / pGateEval / pAcceptR1 / pInitR1	ILL-FORMED (parse errors)
T11/T12 (hs)	the disclosed hs boundary: the computed verdict remains an unevaluated VerifyLineageG(. . .) application; the receipt structure is concrete

7. Python/K correspondence and regression (zero drift)

The mirror (`r2_model.py`) imports the frozen 1G gate UNCHANGED (`authz_model.verify_lineage_g`, backed by the frozen 1D/1E/1F chain) — the same function the K machine evaluates. Two disclosed mirror artifacts are handled explicitly and total on the verdict code set: the frozen Python mirror emits layer-coded verdict names (L29-POLICY for the K `invalid-policy`, ...) and the R2 surface normalizes them (`verdict_k`); the normalization changes no decision. `run_r2_attacks.py` — **87/87 PASS** (`transcripts/attacks/r2-attack-suite.txt`): KX (byte certificate, 11 checks), HONEST (10), INJ (7 — the decisive injection group), VERDICT (20), BIND (15), ALT (2), CFG (4), STALE (3), POOL (4), MAL (2), LEG (3), DET (6).

The cross-layer byte certificate (``transcripts/crosslayer/r2_byte_certificate_k_raw.txt``, K LLVM real-keccak) is reproduced digit-for-digit by Python: `cfgr2root c549b4a7...`, `policycommitr2 e1c78257...`, `ctxdigest0 e5710efa...`, `blockcommit0 8dad13d8...`, `srw3root0 8ef5bb80...`, `certid0 89ffea58...` (= the FROZEN 1G transcript anchor), `certid1 837a5ba4...`, `recchild0 cc07b327...`; verdicts `valid-g/valid-g`.

Zero regression (re-run live, exact):

```
phaselh/python/test_lineage_r1.py 11/11
phaselh/python/test_security_context_r1.py 26/26
phaselh/python/run_attack_matrix.py 30/30
phaselh/python/run_determinism_i.py 15/15
phaselh/python/run_reorg_replay.py 17/17
phaselh/python/run_forkchoice_compare.py 16/16
phaselh-r1/python/run_r1_attacks.py 48/48
```

8. Trust boundaries and scientific claim limits

- Verdict source.** The verdict is the frozen `VerifyLineageG` result; its soundness rests on the frozen gate module (mechanized in 1G). R2 adds no trust to the evidence channel.
- Haskell-backend hash abstraction** (finding 1C-1): digest equalities are symbolic there; the hs demos retain the computed verdict as an unevaluated application (T11). Concrete decisions are the LLVM runs

(real keccak, pinned shim) and the Python mirror.

3. **Keccak collision resistance** remains an assumption for the substituted-digest distinctions (1I/1I-R1 boundary, unchanged).
4. **Execution-evidence authenticity**: that an off-chain executor's trace truthfully describes historical Ethereum execution remains OUTSIDE the model (1F/1H limitation, restated, not implicitly solved by R2).
5. **Level discipline**: R2 is a Level-III protocol-model result. No Ethereum consensus integration, Engine-API change, or client support is claimed.
6. **NOT MECHANIZED residue**: R2-ALL (universal closure) and the two full-pipeline K compositions — each with retained logs and demonstrated concrete substitutes (section 5).
7. **Upgrade path**: deliberately absent; policy upgrade stays a deferred protocol-state transition requiring its own authorization semantics.

9. Completion criteria (handoff checklist)

- [x] the old R1 injection mechanically reproduced (K witness claims + Python + transcripts, §3)
- [x] R2 has no caller-supplied verdict field/parameter (syntax audit + T9 parse transcripts)
- [x] the receipt stores an internally computed `VerifyLineageG` result (mint-shape claims + T1 + KX)
- [x] every gate input is bound to the exact candidate and pinned context/config (`GateInputMatchesBlockR2/§5.2` map, audit §F)
- [x] the old insecure transition is unreachable from the admitted R2 command surface (audit §B + T10a-d)
- [x] invalid candidates cannot commit, including under the original attack (T6-T8, VERDICT/BIND groups, kprove family)
- [x] honest valid candidates still commit (T2/T3, HONEST group, R2-VALID-ACCEPT)
- [x] K proof boundaries and all failed attempts disclosed (§5, logs retained)
- [x] Python/K correspondence and regression suites recorded (§7, 87/87, zero drift)
- [x] report, PDF, and provenance committed
- [x] no frozen refs changed (frozen-ref check in provenance)
- [x] no Phase 1J / novelty audit started

10. Verdict

CLOSED (in-model gate-verdict provenance), with the disclosed residues: R2-ALL and the two full-pipeline K compositions are NOT MECHANIZED on the Haskell backend (logs retained; concrete substitutes demonstrated on LLVM and Python), and the standing 1F/1H boundaries about execution-evidence authenticity, keccak collision resistance, and the Level-III model scope are restated unchanged. The claim "gate provenance solved" rests on the actual

verifier being evaluated, candidate-bound, config-bound, and the only accepted result — not on the removal of the String parameter alone.

Next step (deferred until this phase is reviewed): the Phase 1I-R2 handoff explicitly defers the novelty/prior-art audit and Phase 1J.